VIRAL TRAJECTORY

THE SARS-CoV-2 VIRUS CONTINUES TO EXIST IN OUR MIDST, MUTATING FURIOUSLY FOR SURVIVAL. BUT EVEN AS IT JUMPS FROM HOST TO HOST, IT SEEMS TO BE FAST LOSING ITS DEATHLY STING

HOW COVID IS MUTATING INTO A MORE INFECTIOUS BUT MILDER VIRUS

ALPHA

Infectivity: 30-50 per cent more infectious than the original Wuhan-Hu-1 strain. Emerged in India around Dec. 2020

Severity in India: 1,000 daily deaths at its peak, but soon overthrown by Delta

Common symptoms: Fever, cough, breathing difficulty, bodyache, fatigue

BETA

Infectivity: 50 per cent more infectious than the original. Started doing the rounds in India in Mar. 2021

Severity in India: No data

Common symptoms: Fever, cough, breathing difficulty, bodyache, fatigue

DELTA

Infectivity: 80-90 per cent more infectious than Alpha. India reported first cases in Dec. 2020

Severity in India: 4,000 deaths a day when the wave was at its worst in May 2021

Common symptoms: Fever, cough, fatigue, lung disease, bodyache, sudden hypoxia

OMICRON

Infectivity: Twice as infectious as Delta. Was first detected in India in Dec. 2021

Severity in India: An average of 1,000 deaths daily at its height in Jan. 2022

Common symptoms: Sore throat, fever, bodyache, fatigue

**Sources: CDC, Yale Medicine, MoHFW*

THE OMICRON FAMILY

BA.1.1529

(Original Omicron)

Mutated to become more transmissible than Delta through changes in its spike protein (the part that allows entry into cells)

BA.1.1

Similar mutations as the parent strain but also has a few unique mutations that didn't alter the nature of the virus significantly

BA.2

Studies have shown that BA.2 has a growth advantage over BA.1 due to increased ability to enter cells

BA.2.12.1

In addition to BA.2's increased infectivity, this sub-variant mutated to improve its resistance against human antibodies

BA.2.75.2

A BA.2 sub-variant, it had the most mutations to escape neutralising effects of antibodies

BA.3

There were no unique mutations, but a combination of mutations found in BA.1 and BA.2 sub-variants

BA.4

The first sub-variant to evolve the L452R and F486V mutations in the viral spike protein. This improved its ability to latch onto host cells and skirt some immune responses

BA.4.6

Was relatively uncommon in terms of global reinfections. Had most of the same mutations as BA.4 and a few unique ones that didn't impact the virus behaviour much

BA.5

Mutated to escape antibodies that even BA.2 and its sub-variants hadn't been able to build resistance against. Largely responsible for the wave in the US in August this year. Highly capable of reinfecting those with previous immunity. It is currently dominant around the world

BF.7

A sub-variant of BA.5, it was first found in India on October 18. It has mutations that can dodge human immunity better than any previous sub-variant. It can also bind better to cells and requires less viral load to infect

